

Intro Programming

HW05 — Challenge • Loops • Menu-Driven Program

Due: (your course LMS) • 11:59 PM

HW05: NumberWorkshop

Submit: NumberWorkshop.java

Goal: Write an interactive, menu-based Java program that uses **loops** to compute number properties. You will implement two operations and keep the program running until the user chooses to quit.

DESCRIPTION

In this challenge, you will implement `NumberWorkshop.java`, a console program that repeatedly displays a menu and performs computations selected by the user. Each computation must be implemented using **loops** (not built-in shortcuts).

The user can choose either:

- **(1) Greatest Common Divisor** of two integers
- **(2) Prime Factor Breakdown** of a positive integer
- **(0) Quit** to end the program

OPERATIONS

(1) Greatest Common Divisor (GCD)

The GCD of two integers is the largest positive integer that divides both numbers with no remainder. For example, $\text{gcd}(18, 48) = 6$ and $\text{gcd}(21, 21) = 21$.

- You must compute GCD using a loop (Euclid's algorithm is recommended).
- Handle negative inputs by treating the GCD as the GCD of absolute values.

(2) Prime Factor Breakdown (PFB)

A prime factor breakdown expresses an integer as a product of primes in nondecreasing order. Example: $84 = 2 \times 2 \times 3 \times 7$.

- Input is a positive integer $n \geq 2$.
- Output the prime factors in ascending order, separated by spaces.
- You must use loops to find factors (no precomputed prime lists).

Formatting rule: Print outputs exactly as described below. Autograding is whitespace-sensitive.

PROGRAM BEHAVIOR

On each iteration, print a menu like this (exact text/casing recommended):

```
=== NumberWorkshop ===
1) Compute GCD
2) Prime Factor Breakdown
0) Quit
Choose an option:
```

Then behave as follows:

- If the user enters 1, prompt for two integers and print the GCD result.
- If the user enters 2, prompt for one integer and print its prime factors.
- If the user enters 0, print a goodbye message and terminate.
- For any other input, print an error message and re-display the menu.

Example I/O (illustrative):

```
Choose an option: 1
Enter the first integer: 18
Enter the second integer: 48
GCD(18, 48) = 6

Choose an option: 2
Enter an integer >= 2: 84
Prime factors: 2 2 3 7

Choose an option: 0
Goodbye!
```

IMPLEMENTATION REQUIREMENTS

- File name must be `NumberWorkshop.java` and contain a `public class NumberWorkshop`.
- You must use **loops** for both operations.
- Do not use recursion for the prime factor breakdown.
- Do not use any external libraries beyond standard Java (`java.util` is fine).
- Write at least two helper methods, e.g. `gcd(int a, int b)` and `printPrimeFactors(int n)`.

Suggested helper signatures (you may adapt):

```
static int gcd(int a, int b)
static void printPrimeFactors(int n)
```

TESTING

- Test GCD with: equal numbers, one zero, negatives, co-prime pairs.
 - Test factorization with: prime input (e.g., 97), powers (e.g., 64), and mixed composites (e.g., 420).
 - Test invalid menu options (e.g., 9) and verify the program recovers gracefully.
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SUBMIT

- Submit `NumberWorkshop.java` to your LMS.
 - Make sure your code compiles and runs from the command line.
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